

Problem 1.**Airport data:**

- What is the α if the distribution is power law?
- What is the λ if the distribution is exponential?
- What is $[a, b]$ if the distribution is uniform?
- What is μ and σ if the distribution is normal?

(These can be CDF, PDF, or QQplot)

- Screen shot of the visualization of distribution of data

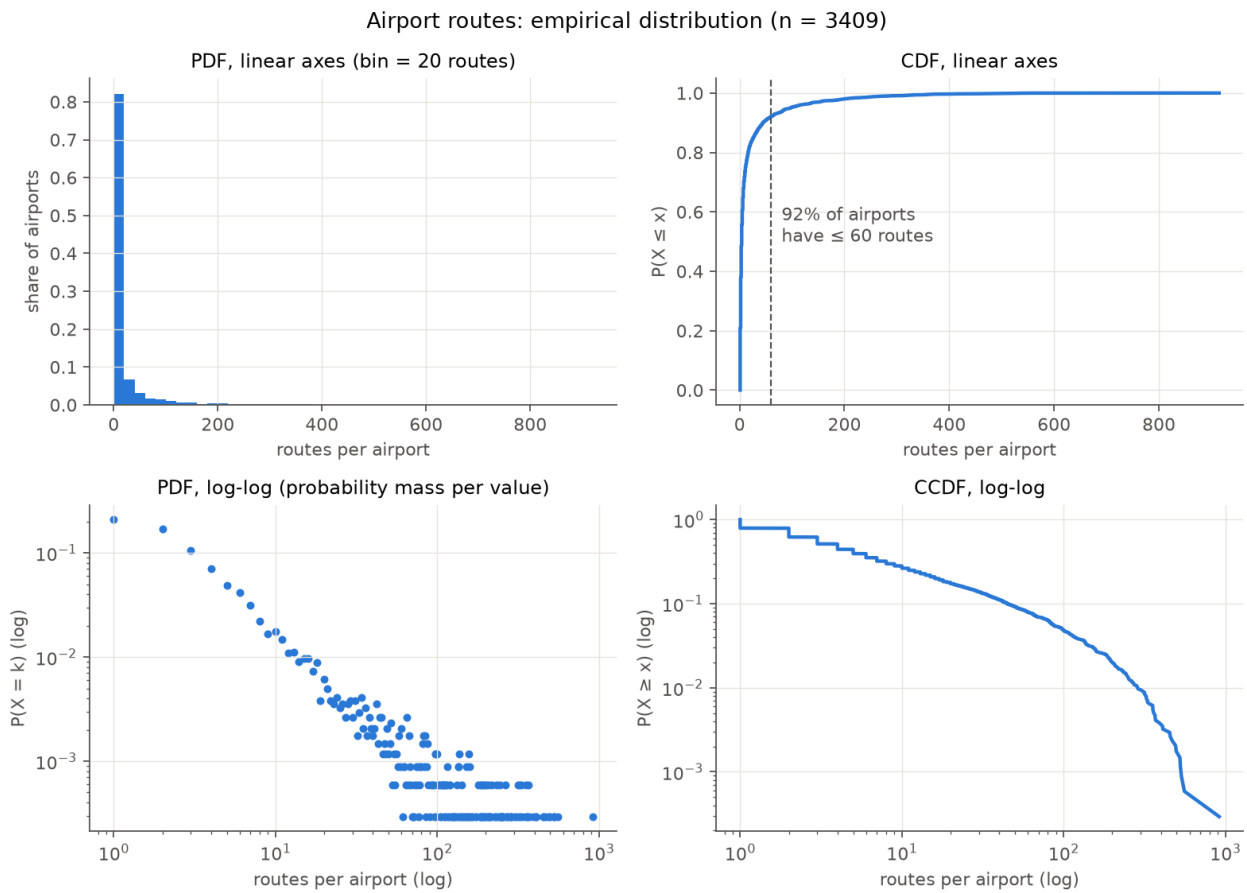


Figure 1: Airport routes: PDF and CDF on linear axes (top), PDF and CCDF on log-log axes (bottom).

- Screen shot of the visualization of distribution of data if it was power law

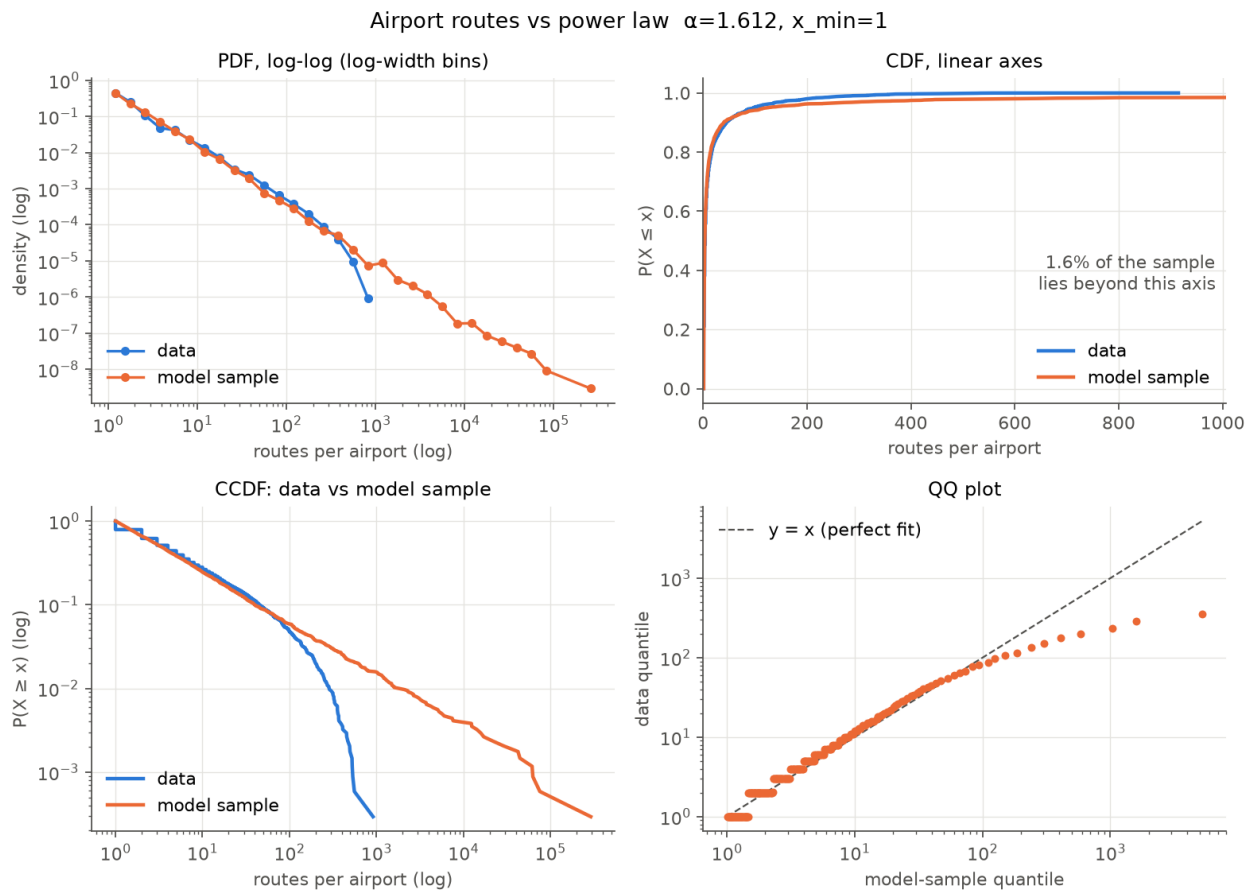


Figure 2: Airport routes vs. samples from the fitted power law.

- Screen shot of the visualization of distribution of data if it was exponential

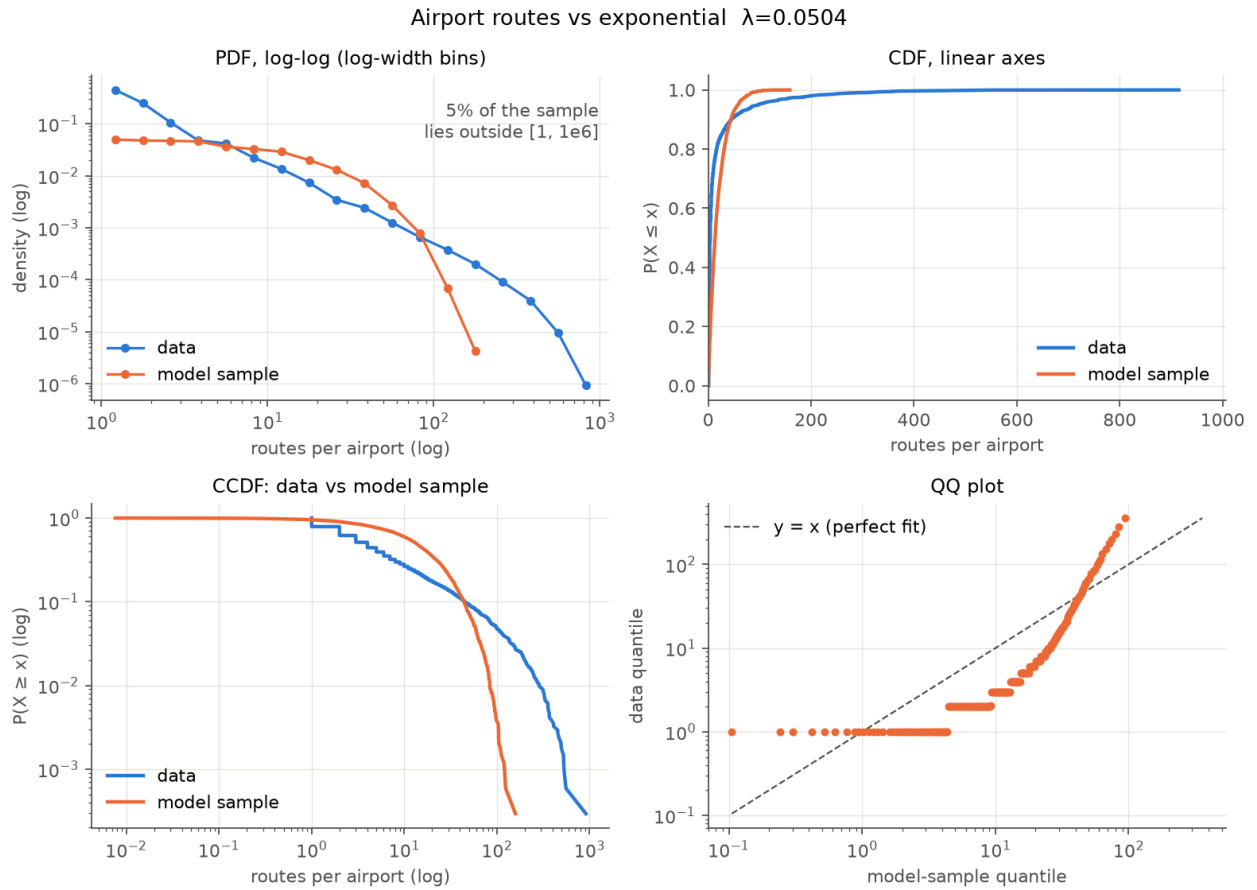


Figure 3: Airport routes vs. samples from the fitted exponential.

- Screen shot of the visualization of distribution of data if it was uniform

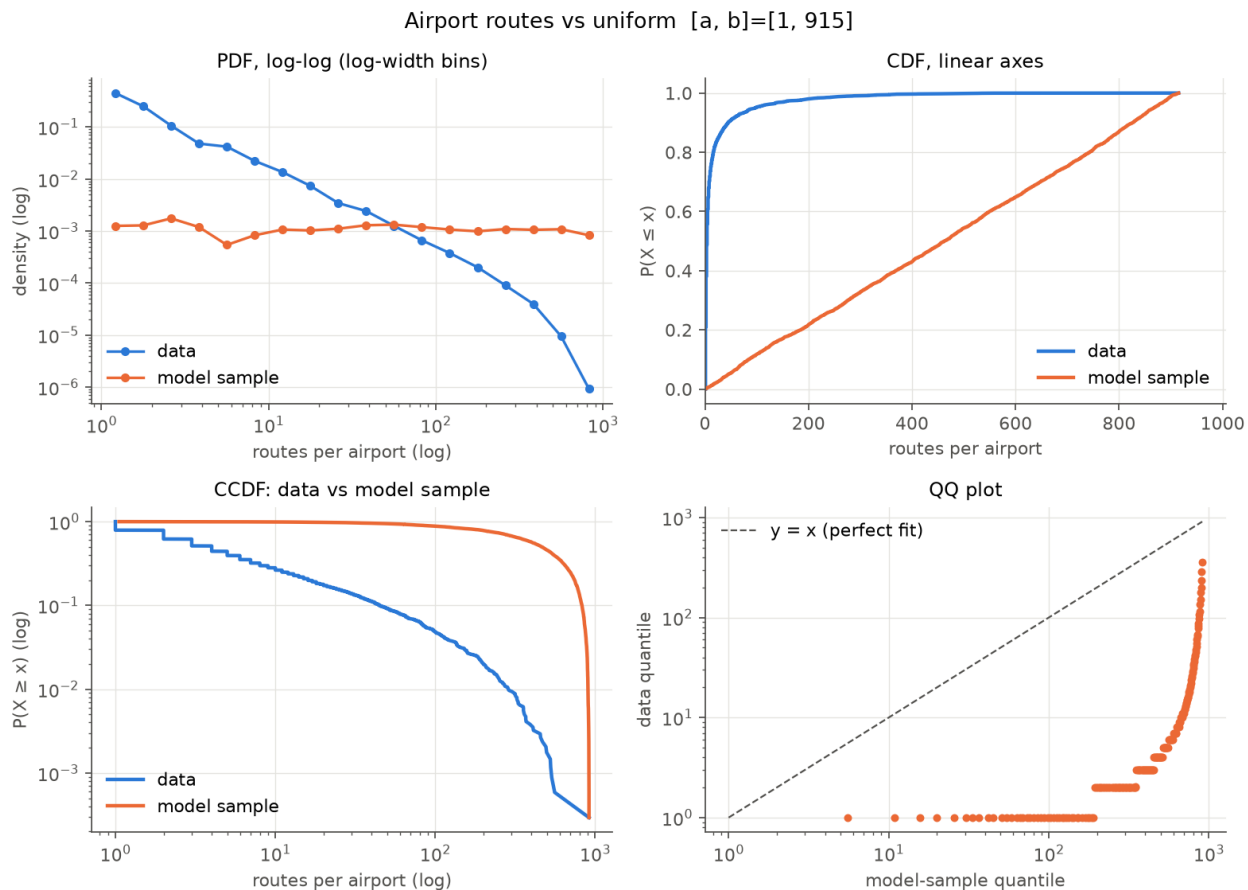


Figure 4: Airport routes vs. samples from the fitted uniform.

- Screen shot of the visualization of distribution of data if it was normal

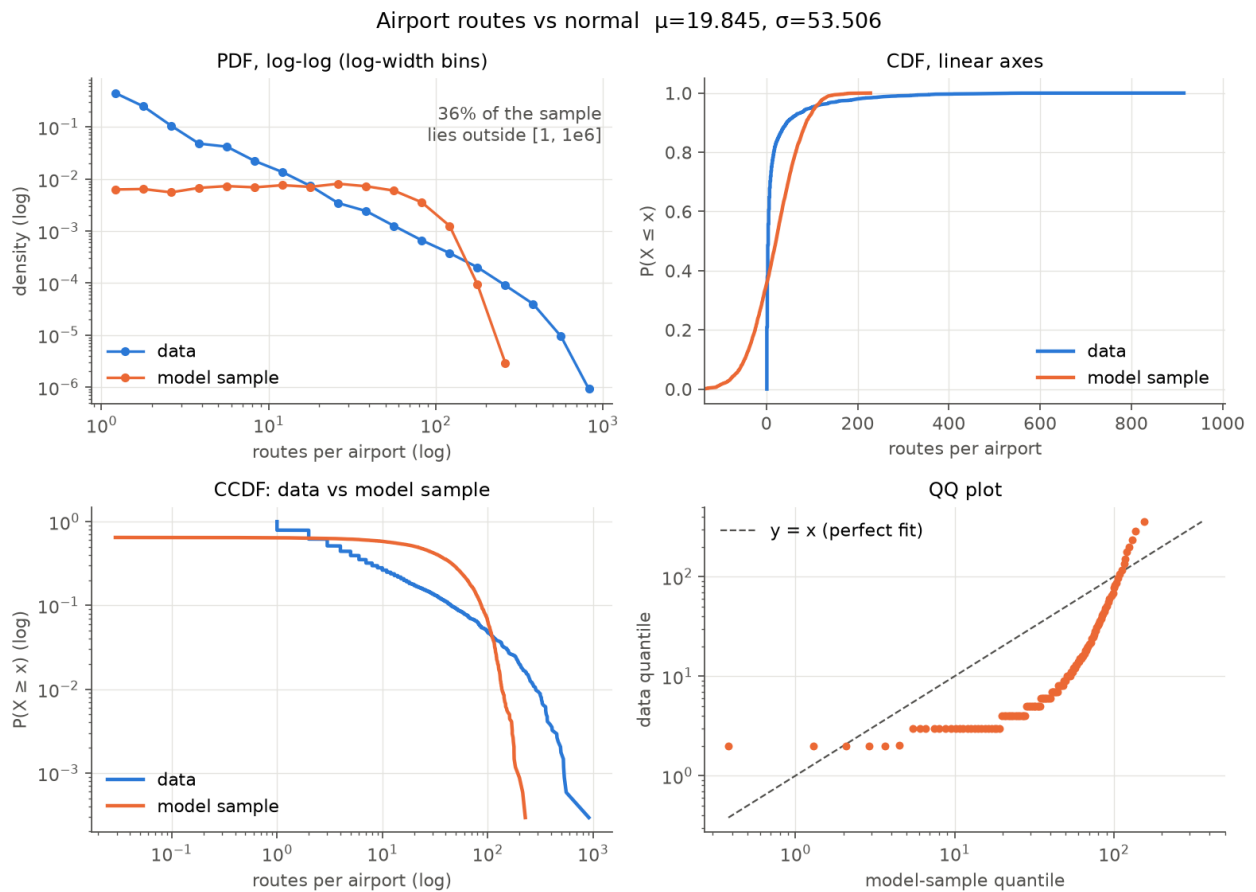


Figure 5: Airport routes vs. samples from the fitted normal.

- What is the distribution of data based on your observation? Discuss

Movie Rating data:

- What is the α if the distribution is power law?
- What is the λ if the distribution is exponential?
- What is $[a, b]$ if the distribution is uniform?
- What is μ and σ if the distribution is normal?

(These can be CDF, PDF, or QQplot)

- Screen shot of the visualization of distribution of data

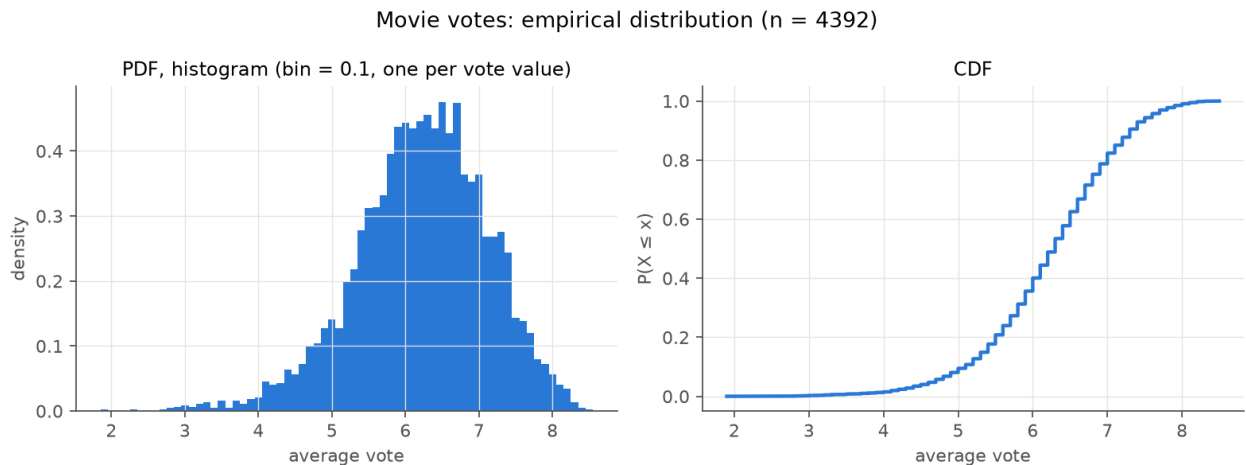


Figure 6: Movie votes: empirical PDF (histogram) and CDF.

- Screen shot of the visualization of distribution of data if it was power law

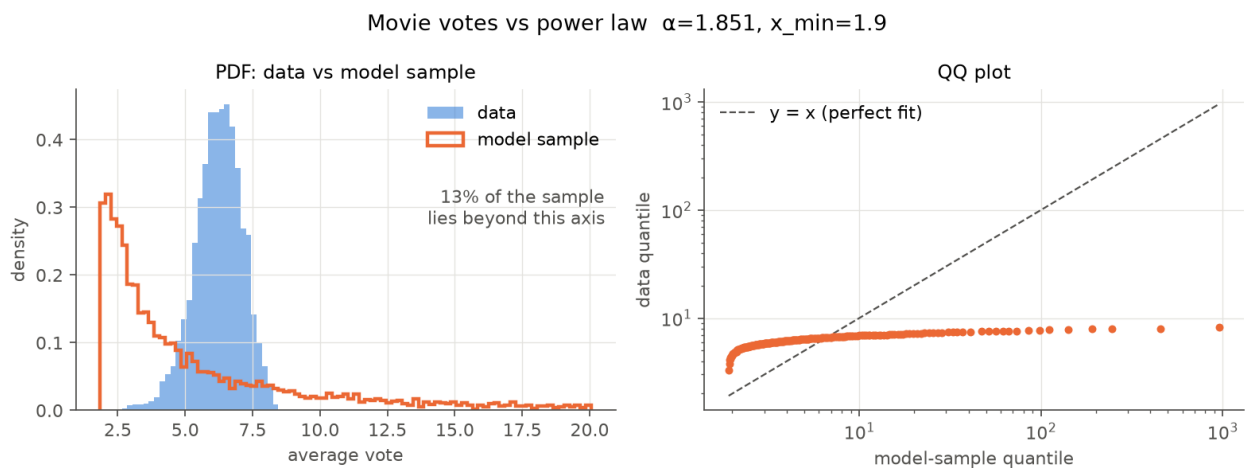


Figure 7: Movie votes vs. samples from the fitted power law.

- Screen shot of the visualization of distribution of data if it was exponential

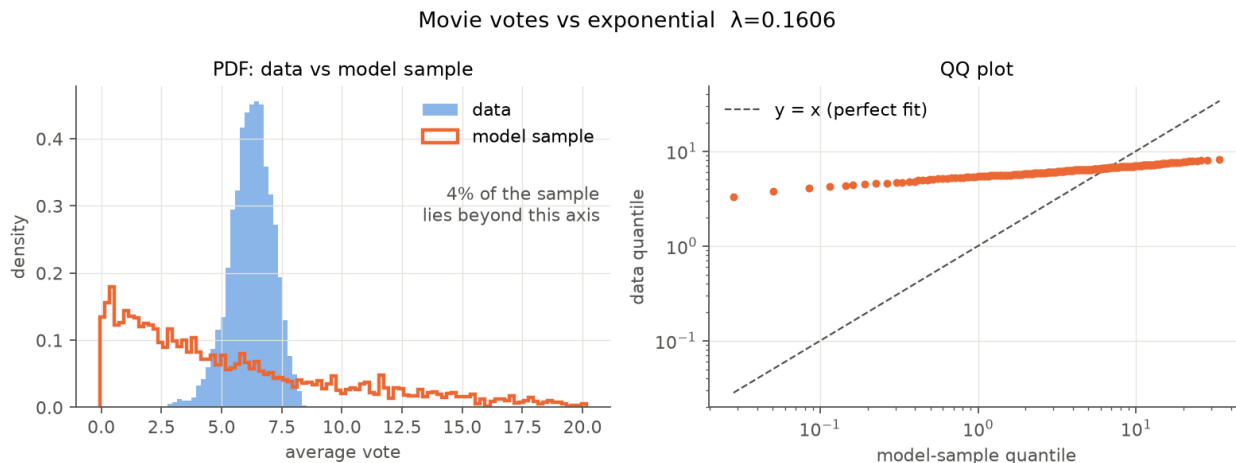


Figure 8: Movie votes vs. samples from the fitted exponential.

- Screen shot of the visualization of distribution of data if it was uniform

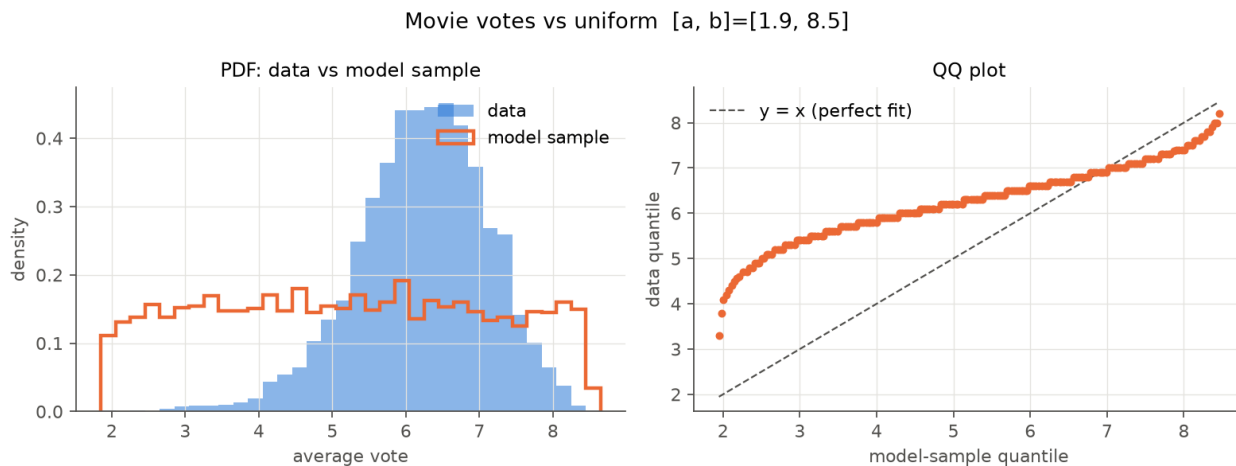


Figure 9: Movie votes vs. samples from the fitted uniform.

- Screen shot of the visualization of distribution of data if it was normal

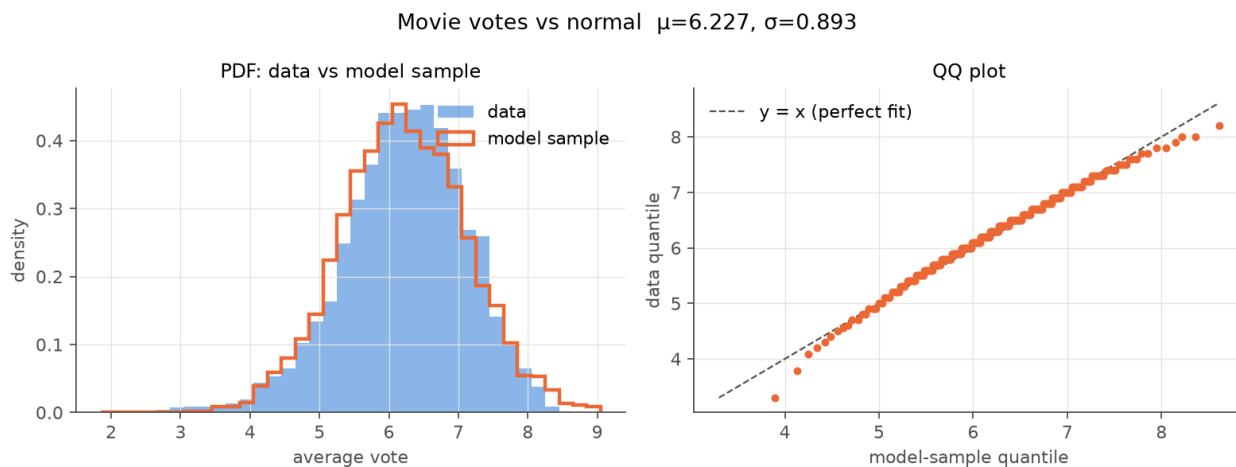


Figure 10: Movie votes vs. samples from the fitted normal.

- What is the distribution of data based on your observation? Discuss

Problem 2.

- Screenshot of distribution/histogram of the age

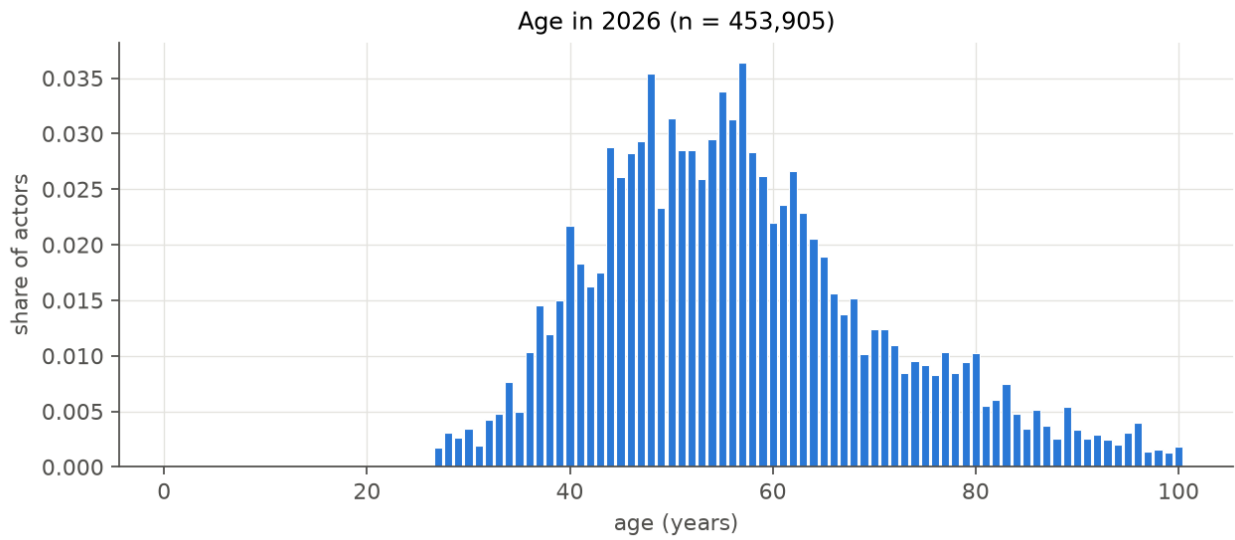


Figure 11: Distribution of age (1 to 99 years).

- Screenshot of distribution/histogram of the first digit of age

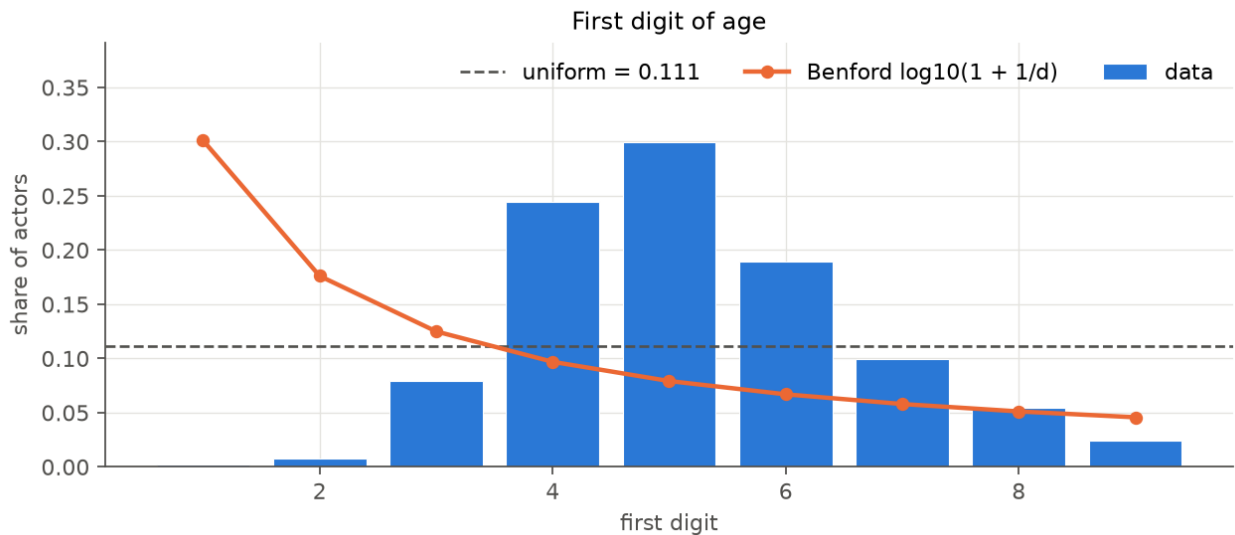


Figure 12: Distribution of the first digit of age.

- Screenshot of distribution/histogram of the last digit of age

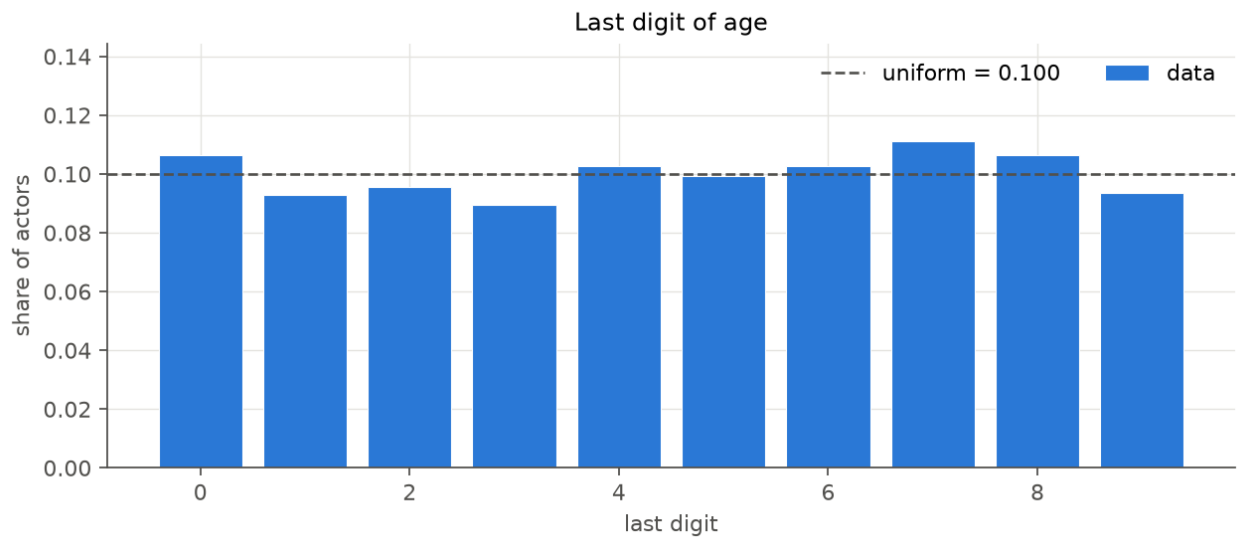


Figure 13: Distribution of the last digit of age.

- Discussion (what are those distribution?) Why do you think it is like this?